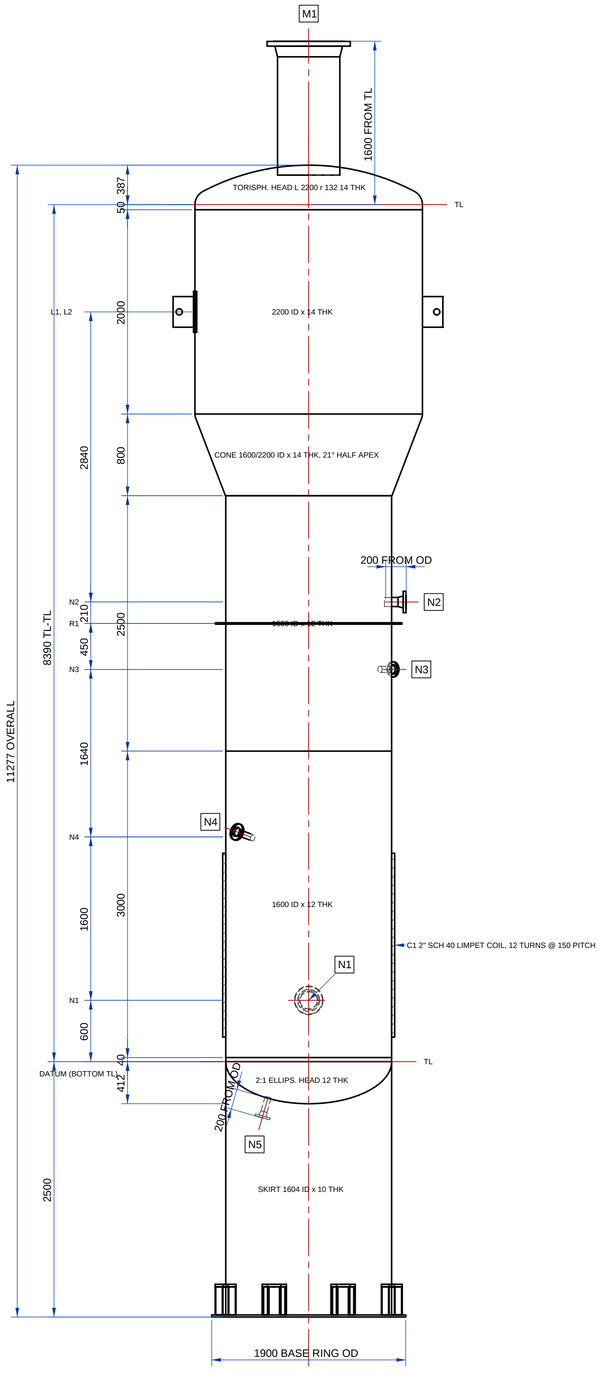
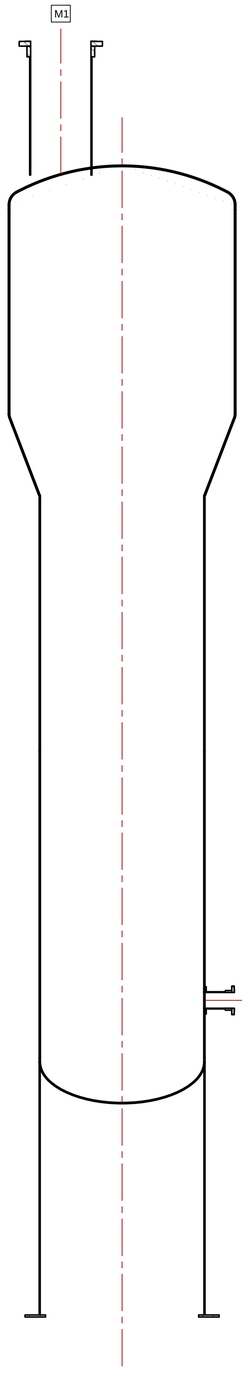


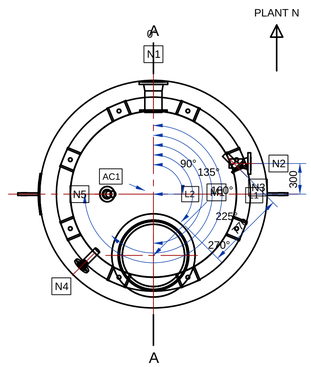
ELEVATION

SCALE 1:30



SECTION A-A

SCALE 1:30



A

PLAN (NOZZLE ORIENTATION)

SCALE 1:30

BILL OF MATERIALS (WEIGHTS: HANDBOOK VALUES WHERE NOT REGENERATED)

ITEM	QTY	MARK	DESCRIPTION	MATERIAL	STD / REF	UNIT kg	TOTAL kg
1	1	E0	SKIRT 10 THK	SA-283 Gr.C		1154.83	1154.83
2	1	E1	BOTTOM HEAD	SA-516 Gr.70		284.43	284.43
3	1	E2	CYLINDER 12 THK	SA-516 Gr.70		1431.16	1431.16
4	1	E3	CYLINDER 12 THK	SA-516 Gr.70		1192.63	1192.63
5	1	E4	CONE 14 THK	SA-516 Gr.70		564.61	564.61
6	1	E5	CYLINDER 14 THK	SA-516 Gr.70		1528.82	1528.82
7	1	E6	HEAD 14 THK	SA-516 Gr.70		538.46	538.46
8	1	N1	NOZZLE N1 NECK 168.3 × 7.11 WITH PAD	SA-106 Gr.B		5.62	5.62
9	1	N1	FLANGE WN OD 280	SA-106 Gr.B		11.16	11.16
10	1	N2	NOZZLE N2 NECK 88.9 × 7.62	SA-516 Gr.70		1.83	1.83
11	1	N2	FLANGE WN OD 210	SA-516 Gr.70		7.56	7.56
12	1	N3	NOZZLE N3 NECK 60.3 × 5.54	SA-516 Gr.70		0.64	0.64
13	1	N3	FLANGE WN OD 150	SA-516 Gr.70		2.74	2.74
14	1	N4	NOZZLE N4 NECK 60.3 × 5.5	SA-516 Gr.70		0.97	0.97
15	1	N4	FLANGE WN OD 165	SA-516 Gr.70		3.69	3.69
16	1	N5	NOZZLE N5 NECK 60.3 × 5.54	SA-516 Gr.70		1.02	1.02
17	1	N5	FLANGE WN OD 150	SA-516 Gr.70		2.74	2.74
18	1	M1	NOZZLE M1 NECK 610 × 9.53	SA-516 Gr.70		161.35	161.35
19	1	M1	FLANGE WN OD 815	SA-516 Gr.70		120.53	120.53
20	1	L1	LIFTING LUG 25 THK	SA-516 Gr.70	EIL 7-12-0024 R5	11.22	11.22
21	1	L2	LIFTING LUG PAD 14 THK	SA-516 Gr.70	EIL 7-12-0024 R5	17.69	17.69
22	1	L2	LIFTING LUG 25 THK	SA-516 Gr.70	EIL 7-12-0024 R5	11.22	11.22
23	1	R1	STIFFENING RING BAR 100X12	SA-516 Gr.70		51.02	51.02
24	1	C1	LIMPET COIL NPS 2 SCH 40 HALF PIPE (SOLID)	SA-516 Gr.70		1373.09	1373.09
25	16	AC1	ANCHOR CHAIR GUSSET 16 THK	SA-516 Gr.70		4.77	76.32
26	8	AC1	ANCHOR CHAIR TOP PLATE 25 THK	SA-516 Gr.70		5.22	41.76
			TOTAL (EMPTY)				8597.11

DESIGN DATA

ITEM	VALUE
DESIGN CODE	ASME VIII-1 2025
MAWP INTERNAL	1000 kPa(g) @ 150 °C
MAWP EXTERNAL	103 kPa(g) @ 150 °C
DESIGN TEMPERATURE	150 °C
MDMT	-29 °C
HYDROTEST PRESSURE	1300 kPa(g)
CORROSION ALLOWANCE	3 mm
JOINT EFFICIENCY	0.85
RADIOGRAPHY	Spot
PWHT	NO
WIND / SEISMIC	IS 875 Part 3 / IS 1893
CAPACITY	21.5 m³
MATERIAL (SHELL)	SA-516 Gr.70
WEIGHT EMPTY	8597.1 kg
WEIGHT OPERATING	8597.1 kg
WEIGHT TEST	31231.7 kg
SERIAL / YEAR	OET-0001 / 2026

NOZZLE SCHEDULE

MARK	QTY	SIZE	SCH/THK	RATING	FACING	TYPE	ELEV.	AZIM.	PROJ.	SERVICE	REMARKS
N1	1	6"	STD	B16.5 CL 150	RF	WN	600	0°	1100 CL	Inlet	PAD 50 x 12
N2	1	3"	XS	B16.5 CL 300	RF	WN	4500	90°	200 OD	Level gauge	HILLSIDE 300
N3	1	2"	80	B16.5 CL 150	RF	WN	3840	135°	150 OD	Tangential inlet	TANGENTIAL
N4	1	2"	5.5 THK	B16.5 CL 300	RTJ	WN	2200	225°	200 OD	Sample	TILTED 15°, FORGED
N5	1	2"	XS	B16.5 CL 150	RF	WN	ON E1	270°	200 OD	Drain	NORMAL AT R 400
M1	1	24"	STD	B16.5 CL 150	RF	WN	ON E6	180°	1600 CL	Manway	AXIAL AT R 600, MANWAY, DAVIT

NOTES

	NOTE
1.	ALL DIMENSIONS IN mm UNLESS NOTED.
2.	ELEVATIONS FROM THE DATUM (BOTTOM TL).
3.	NOZZLE ORIENTATION: 0° = PLANT NORTH, CLOCKWISE LOOKING DOWN.
4.	WELDS ARE NOT MODELLED; WELD DETAILS PER THE FABRICATION SPECIFICATION.
5.	NOZZLE ORIENTATION LOOKING DOWN, 0° = PLANT NORTH, CLOCKWISE.
6.	CHECK: ATTACHMENT `L1`: NO PRESET `LL-25` FOR EIL 7-12-0024:5 IN THE TABLES (IMPORT IT); ITS OWN DIMENSIONS AR..
7.	CHECK: ATTACHMENT `L2`: NO PRESET `LL-25` FOR EIL 7-12-0024:5 IN THE TABLES (IMPORT IT); ITS OWN DIMENSIONS AR..

REVISIONS

REV	DATE	BY	DESCRIPTION
0	2026-11-08	SSP	ISSUED FOR BETA

OPEN ENGINEERING TOOLS

PROJECT: BETA REFERENCE SET

TITLE: REFERENCE VESSEL A (VERTICAL, SKIRT-SUPPORTED)

DRAWING NO.	REV	SIZE
V-101-GA-001	0	A1

DRAWN	CHECKED	DATE	SCALE	SHEET
SSP		2026-11-08	1:30	1 OF 1